

★ PRIMER · UNIT 7

Before the Lecture: Asking the Right Question

Unit 7 · Trees and Ensembles

Watch this first if impurity and weighted averages are new

Twenty minutes of groundwork — how a tree picks a question, what impurity measures, and why a crowd of weak models beats one clever one.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Gini and weighted averages are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 7 Sits

The course builds one pipeline; here is this unit's stone

Unit 6

done

The widest road — support
vectors.

Unit 7

you are here

One tree, then a forest of them.

Unit 8

next

Neurons, layers and
backpropagation.

This is the first model that needs no scaling and no straight line at all.

What a Tree Actually Does

A tree is a sequence of yes-or-no questions, each chosen to make the groups purer than before.

How it splits

- Try every question.
- Score each by purity gained.
- Keep the best, then repeat.

Why it is friendly

- No scaling needed.
- Handles non-linear shapes.
- You can read the rules.

Its weakness is the flip side of its strength — left alone, it memorises everything.

The Words the Lecture Assumes

Five terms, and the forest section opens up

Word	What it means	Here
Node	one question in the tree	is income above 40?
Leaf	the end — a verdict	default, or not
Impurity	how mixed a group is	0 means all one class
Gini	one way to score impurity	the lecture's choice
Depth	how many questions deep	the main overfitting knob

Impurity zero means the group is pure. That is the target every split aims at.

Two Ways to Build a Crowd

Both beat one tree, and they disagree about how

Method	How trees are built	What it fixes
Bagging	in parallel, on random samples	variance — the memorising
Random Forest	bagging, plus random features	variance, harder
Boosting	in sequence, each fixing the last	bias — the systematic misses

XGBoost is boosting. It is the model that keeps winning competitions, and unit 7 says why.

Why a Crowd Beats One Expert

Many weak models, each wrong in a different direction, average their errors away.

One deep tree

- Fits the training set perfectly.
- Every quirk is memorised.
- New data breaks it.

A hundred shallow ones

- Each sees a different sample.
- Their mistakes disagree.
- The average is stable.

The trees must disagree. That is why the forest randomises both rows and columns.

Gini by Hand

Six loans, one question — is the split worth making?

Group	Default	Repaid	Gini
Before split	3	3	0.500
Income low	3	1	0.375
Income high	0	2	0.000
Weighted after	—	—	0.250

$$\text{Gini} = 1 - p_1^2 - p_2^2. \text{ Weighted after} = (4/6)(0.375) + (2/6)(0) = 0.250.$$

Reading That Split

One number decides whether the question was worth asking

The gain

- Before: 0.500.
- After: 0.250.
- Gain is 0.250.

What it means

- The groups got much purer.
- High income is now pure.
- The tree keeps this question.

The tree tries every question and keeps the one with the largest gain

Why One Tree Memorises

Left unpruned, a tree will happily give every row its own leaf

Unlimited depth

- Keep splitting until pure.
- Every row gets a leaf.
- Training accuracy: 100%.

On new data

- The rules were about noise.
- Accuracy collapses.
- Classic high variance.

Depth, minimum leaf size and the number of features are the knobs that stop this.

How Boosting Fixes Mistakes

Three moves, repeated a few hundred times

1 · Fit

a weak tree

A shallow tree, deliberately not very good on its own.

2 · Measure

what it missed

Look at the residuals — where the crowd is still wrong.

3 · Add

a tree for that

Fit the next tree to the mistakes, and add it in.

Each tree is small. What makes boosting strong is that each one is aimed at the remaining error.

Three Ways to Rank a Feature

They disagree, and the lecture shows you when

Method	How it scores	Weakness
MDI	total impurity it removed	favours many-valued features
MDA	damage done by shuffling it	slow, and noisy
SFI	how it does on its own	misses interactions

Unit 14 replaces all three with SHAP, and explains why they were never quite honest.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
p	p	the proportion of one class in a group
Gini	gee-nee	1 minus the sum of squared proportions
n_estimators	n estimators	how many trees are in the crowd
max_depth	max depth	how many questions deep a tree may go

The last two are not maths — they are the two knobs you will actually turn.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>DecisionTreeClassifier(max_depth=3)</code>	one shallow, readable tree
<code>RandomForestClassifier(n_estimators=100)</code>	a hundred trees, voting
<code>XGBClassifier()</code>	boosting — trees that fix each other
<code>model.feature_importances_</code>	MDI, with all the caveats above
<code>IsolationForest()</code>	the unit's anomaly detector

Try `max_depth=2` and print the tree. Being able to read the rules is the point.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 8

1

"Feature importance tells you what causes what"

It tells you what the model leaned on. Unit 14 spends a whole part on that difference.

2

"More trees can overfit"

In a random forest, more trees only stabilise. Depth overfits; count does not.

3

"Trees need their features scaled"

They do not. A tree only asks whether a value is above a threshold, so units cancel.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 A group is 3 default, 3 repaid. What is its Gini?

0.500

2 Which builds trees in sequence, each fixing the last?

boosting

3 Which knob makes a single tree memorise?

depth

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It builds one tree, then breaks it

Gini by hand, then the memorising weakness and the knobs that tame it.

2

It assembles the crowd

Bagging, random forest and boosting, with the error melting away step by step.

3

It ends in code

The problem set scores a split by hand, takes a forest vote, and isolates an outlier.

Next: Unit 7 lecture — Trees and Ensembles